

RAG

1.Vectors (Embeddings)

- These are **numbers that represent meaning of text, image, audio, etc.**
- Example:
 - “cat” → [0.12, 0.55, 0.91, ...]
 - “dog” → [0.10, 0.57, 0.89, ...]

Similar meanings → similar vectors.

Semantic Search

2.Original Data

- The real text, image, or document is stored.
- Example:
 - Sentence: *"Dogs are friendly animals"*
 - PDF, image, product info, etc.

3.Metadata

- Extra info about data.
- Example:
 - author = "VJ"
 - date = "2026"
 - category = "AI notes"

Vector DB = Vector + Original Data + Metadata

4.ID (Unique Identifier)

- A unique number or string for each data item.
- Vector databases use a **hash map / hash table** to store and find data by ID fast.
- Because hash table gives **O(1) lookup time** (very fast).
- Exact Search
- Example:
 - id = 101
 - id = "doc_001"

Used to quickly find ,delete or update data.

RAG

RAG = Retrieval Augmented Generation

AI first searches data, then generates the answer.

Simple Example

1. You ask: “**What is my company policy?**”
2. AI **retrieves documents from database** (vector DB).
3. AI **generates answer using those documents**.

RAG is a system where AI retrieves relevant data first and then generates the answer.

Why RAG is Introduced (Very Short & Simple)

Because AI models don't know your private or latest data.

Problems RAG Solves

1.Outdated Knowledge Problem

- LLM is trained on old data.
- RAG fetches latest data from database.

2.Hallucination Problem

- AI sometimes makes fake answers.
- RAG gives real documents → reduces fake answers.

3.Private Data Problem

- Company data is not in ChatGPT training.
- RAG searches your internal files.

RAG in Real Life (Girlfriend WhatsApp Example)

Without RAG

You ask AI:

“When did my girlfriend say I love you last time?”

AI does not have your chat data.

So it **guesses** and may give a wrong answer.

With RAG

AI first **searches your WhatsApp chats**.

It finds the real message.

Then it **answers correctly**.

RAG means: Search real data first, then answer.

“RAG helps AI use real documents instead of guessing.”

it takes your relevant data or you can say that chat

Real-World Example: 1 Lakh WhatsApp Chats

Imagine you have **100,000 WhatsApp messages with your girlfriend.**

Without RAG (Normal AI)

You ask AI:

“What did she say about marriage?”

AI:

- Cannot read 1 lakh chats
- Will guess
- Answer may be wrong (fake)

With RAG (Smart AI)

Step 1: Store Chats in Vector Database

All 1 lakh chats are stored with vectors.

Step 2: Retrieve

You ask:

“What did she say about marriage?”

RAG system searches 1 lakh chats and finds only **relevant messages** like:

- “I want to marry after 2 years”
- “Marriage is important for me”

Step 3: Generate Answer

AI reads only those messages and replies:

“She said she wants to marry after 2 years.”

RAG helps AI find important messages from 1 lakh chats and answer correctly.

“RAG retrieves relevant data from large datasets like 100,000 chats and then generates an accurate answer.”

How Vector DB Stores 1 Lakh Chats (Chunk Example)

When you have **100,000 messages**, we don't store everything as one big text.
We **break it into chunks**.

Example Chunking

Imagine messages:

- Message 0 to 1000 → Chunk 1
- Message 1001 to 2000 → Chunk 2
- Message 2001 to 3000 → Chunk 3
- ...
- Message 99001 to 100000 → Chunk 100

What happens for each chunk?

For every chunk:

1. Convert text → **Vector (numbers)**
2. Store in Vector DB
3. Give an **ID**
4. Store metadata (chat date, sender, corresponding data according to vector etc.)

Vector DB stores chats in small chunks so search is fast and accurate.

Why chunking needed?

- AI cannot read 1 lakh messages at once
- Chunking helps find only useful parts

LangChain

LangChain is a tool to build smart AI apps using LLMs (like ChatGPT).

Problem (LLM alone is dumb in real world)

LLM only:

- Knows **training data only**
- Cannot **read your files, DB, API, PDFs, chats**
- Cannot remember long history
- Cannot connect tools easily

Example:

You: “Answer from my company PDF”

LLM: “I don’t have your PDF”

What LangChain Does (Solution)

LangChain helps you:

- Connect LLM with **database, files, APIs**
- Store and retrieve memory (vector DB)
- Build chatbots, agents, tools
- Create RAG systems (search + AI answer)

It acts like **glue between AI and real world data**.

Why We Need LangChain (Simple)

Without LangChain

LLM = brain without hands

With LangChain

LLM = brain + tools + memory

Real World Example (Girlfriend WhatsApp Chat)

Imagine:

- You have **1 lakh WhatsApp messages**
- You ask AI: “What did she say about marriage?”

LLM alone:

Cannot read your chat history

LangChain + Vector DB:

Searches chats → finds relevant messages → gives answer

LangChain is a framework to connect AI with data, tools, memory, and workflows.



Pinecone is a vector database used to store and search AI embeddings (numbers that represent text, images, etc.) very fast.